

Work Energy and Power

SHORT NOTES

Work Done by constant Force

$$W = \vec{F} \cdot \vec{S}$$

Work Done by Multiple Forces

$$W = [\Sigma \vec{F}] \cdot \vec{S}$$

$$W = \vec{F}_1 \cdot \vec{S} + \vec{F}_2 \cdot \vec{S} + \vec{F}_3 \cdot \vec{S} + \dots (\because \Sigma \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots)$$

$$\text{or } W = W_1 + W_2 + W_3 + \dots$$

Work Done by Variable Force

$$W = \int dW = \int \vec{F} \cdot d\vec{s}$$

- Area under the force and displacement curve on displacement axis gives work done.

Relation between Momentum and Kinetic Energy

$$K = \frac{p^2}{2m} \text{ and } p = \sqrt{2mK}; p = \text{linear momentum}$$

Potential Energy

$$U_f - U_i = - \int_{r_i}^{r_f} \vec{F} \cdot d\vec{r} = -W_{\text{Conservative}}$$

Conservative Force

$$\vec{F} = -\frac{dU}{dr} \hat{r}, \quad \vec{F} = -\frac{\partial U}{\partial x} \hat{i} - \frac{\partial U}{\partial y} \hat{j} - \frac{\partial U}{\partial z} \hat{k}$$

Equilibrium Conditions

- Stable Equilibrium

$$F(r) = -\frac{\partial U}{\partial r} = 0; \text{ therefore } \frac{\partial F}{\partial r} < 0; \text{ and } \frac{\partial^2 U}{\partial r^2} > 0$$

- Unstable Equilibrium

$$F(r) = -\frac{\partial U}{\partial r} = 0; \text{ therefore } \frac{\partial F}{\partial r} > 0; \text{ and } \frac{\partial^2 U}{\partial r^2} < 0$$

- Neutral Equilibrium

$$F(r) = -\frac{\partial U}{\partial r} = 0; \text{ therefore } \frac{\partial F}{\partial r} = 0; \text{ and } \frac{\partial^2 U}{\partial r^2} = 0$$

Work-Energy Theorem

$$W_{\text{All}} = \Delta K \Rightarrow W_C + W_{NC} + W_{\text{Ext}} + W_{\text{Pseudo}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Law of Conservation of Mechanical Energy

If the net external force acting on a system is zero, then the mechanical energy is conserved.

$$K_f + U_f = K_i + U_i$$

Power

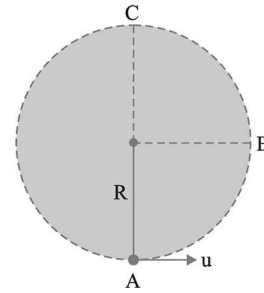
The average power delivered by an agent is given by

$$P_{\text{avg}} = \frac{W}{t}$$

The instantaneous average power delivered by an agent is given by

$$P_{\text{inst}} = \frac{d\vec{F} \cdot \vec{S}}{dt} = \vec{F} \cdot \frac{d\vec{S}}{dt} = \vec{F} \cdot \vec{v}$$

Circular Motion in Vertical Plane



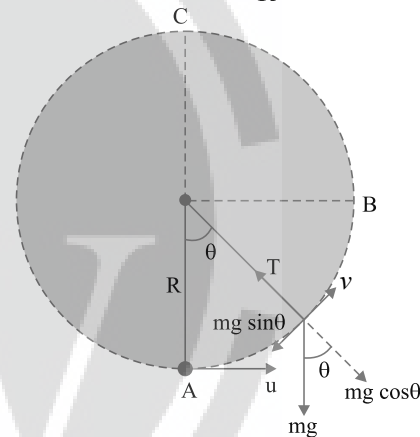
A. Condition to complete vertical circle $u \geq \sqrt{5gR}$

If $u = \sqrt{5gR}$ then Tension at C is equal to 0 and tension at A is equal to 6 mg.

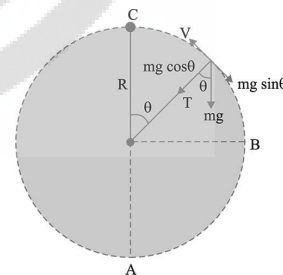
$$\text{Velocity at B: } v_B = \sqrt{3gR}$$

$$\text{Velocity at C: } v_C = \sqrt{gR}$$

$$\text{From A to B: } T = mg \cos \theta + \frac{mv^2}{R}$$



$$\text{From B to C: } T = \frac{mv^2}{R} - mg \cos \theta$$



B. Condition for pendulum motion (oscillating condition)

$$u \leq \sqrt{2gR} \text{ (in between A to B)}$$

Velocity can be zero but T never be zero between A and B.

$$\text{Because T is given by } T = mg \cos \theta + \frac{mv^2}{R}.$$